FEASIBILITY STUDY

SkillLens

Feasibility Study

Project	SkillLens — Skill-to-Employment Intelligence Platform
Institute	Swami Vivekanand College of Engineering, Indore
Mentor	Ms. Amisha Jain
Authors	Teena Devray · Rohit Solanki · Vishal Panwar
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1. Purpose of This Document

This feasibility study evaluates whether the SkillLens project is viable across three dimensions: technical, economic, and operational. Its purpose is to confirm — before any development begins — that the project can be realistically completed within the team's skills, time, and resource constraints, and that the resulting system will be genuinely useful to its target users.

2. Technical Feasibility

Technical feasibility assesses whether the team has the tools, skills, and infrastructure to build the proposed system.

2.1 Team Skills Assessment

Technology	Required Proficiency	Team Current Level	Gap
React.js	Component design, hooks, routing, state management	Intermediate	Minimal — minor revision needed
Node.js + Express.js	REST API design, middleware, file handling	Intermediate	Minimal
MongoDB + Mongoose	Schema design, CRUD, indexing, aggregation	Beginner–Intermediate	Moderate — aggregation needs study
JWT + bcrypt	Auth flow, token validation	Beginner	Low — well-documented, straightforward
OpenAI API	Prompt engineering, JSON response parsing	Beginner	Low — API is well-documented with examples
pdf-parse	PDF text extraction	No prior experience	Very low — single-function library
Tailwind CSS	Responsive utility-class styling	Beginner–Intermediate	Minimal
Recharts	Line charts, bar charts	No prior experience	Very low — declarative, simple API

Overall technical feasibility: HIGH. All required technologies are well-documented, have large community support, and are learnable within the project timeline. No component requires proprietary tools, specialised hardware, or skills outside the reach of a final-year CS student.

2.2 Infrastructure Requirements

Requirement	Solution	Cost	Feasibility
Frontend hosting	Vercel free tier	Free	Confirmed available
Backend hosting	Render free tier	Free	Confirmed available
Database	MongoDB Atlas free tier (512 MB)	Free	Sufficient for project scale
AI evaluation	OpenAI API — gpt-3.5-turbo	~₹0.12 per 1K tokens	Free credits available; low usage expected
Version control	GitHub free tier	Free	Already in use
Development environment	VS Code, Postman, Node.js	Free	Already installed

2.3 Technical Risks and Mitigations

Risk	Likelihood	Impact	Mitigation
OpenAI API unavailable or rate-limited	Low	Medium	System designed to degrade gracefully — interview module works without AI scoring
pdf-parse fails on scanned PDFs	Medium	Low	Scope clearly limited to text-based PDFs; user is notified if extraction

			fails
MongoDB Atlas free tier storage limit	Low	Low	Project-scale data (hundreds of users, lightweight documents) is well within 512 MB
Render free tier cold start delay	High	Low	Expected and acceptable for a project demonstration context

3. Economic Feasibility

Economic feasibility evaluates the cost of building and running the system against the value it delivers.

3.1 Development Cost

Cost Item	Details	Estimated Cost
Software licences	All tools are free and open-source (VS Code, Node.js, MongoDB, React)	₹0
Hosting — development	localhost development environment	₹0
Hosting — production	Vercel + Render free tiers	₹0
Database	MongoDB Atlas free tier	₹0
OpenAI API	gpt-3.5-turbo; estimated 500 test calls × ~200 tokens = 100K tokens total	~₹120
Domain name (optional)	For public demonstration URL	~₹800/year
Total estimated cost		< ₹1,000

3.2 Value Delivered vs. Cost

The platform delivers a service that currently does not exist in this integrated form for Indian students. The nearest comparable paid tools (premium LinkedIn, paid resume services, paid mock interview platforms) cost students ₹500–₹2,000 per month. SkillLens delivers equivalent or superior functionality at zero cost to the student, making the economic case extremely favourable.

3.3 Monetisation Path (Post-Project)

Revenue Stream	Model	Estimated Revenue
Student Pro plan	₹99/month or ₹599/6 months — unlimited access	Scales with user base
College institutional plan	₹15,000/year per college — batch analytics dashboard	Primary B2B revenue stream
Contextual job listings	Companies pay ₹5,000+ to show postings to high-match students only	Secondary revenue

Economic feasibility: HIGH. Development cost is under ₹1,000. The platform has a clear, realistic monetisation path. The free tier ensures adoption without financial barrier.

4. Operational Feasibility

Operational feasibility assesses whether the system will actually be used and whether the team can maintain it.

4.1 User Adoption Likelihood

Factor	Assessment
Problem relevance	Campus placement is a high-stakes, time-pressured event for every

	engineering student — the motivation to use a preparation tool is extremely high.
Ease of use	The platform is designed for non-technical self-use: upload a PDF, take a quiz, see results. No configuration, no manual input beyond the resume.
Free access	The core features are free. Price is never a barrier to adoption.
Immediate value	The company match percentage and gap analysis deliver actionable insight within minutes of first use — no long onboarding required.
Tier 2/3 city relevance	Explicitly designed for students who may lack access to coaching classes or paid preparation tools — a large, underserved market.

4.2 Maintenance Feasibility

The two data assets that require ongoing maintenance are the company profiles JSON and the resource library JSON. Both are plain text files editable without code changes. Updating a company's required skills or adding a new YouTube link requires no redeployment — files can be updated and the backend restarted in under 5 minutes.

The quiz question bank is static and requires no maintenance unless skill requirements change significantly — unlikely within a 1–2 year horizon.

4.3 Timeline Feasibility

The project is scoped for 15 development days (approximately 60 hours of focused work). Based on the team's current skill level and the modular architecture, this timeline is achievable with the following conditions:

- Backend is built first (Days 1–7), providing a stable API for frontend integration
- Company profiles and resource library JSON are built in parallel during backend development
- Frontend development proceeds module by module, not all at once
- OpenAI integration is isolated to the interview module and built last, to avoid API cost during early testing

5. Feasibility Summary

Dimension	Verdict	Key Reason
Technical	FEASIBLE	All technologies are known, free, and well-documented. No specialised hardware or expertise required.
Economic	FEASIBLE	Total development cost under ₹1,000. Clear monetisation path post-project.
Operational	FEASIBLE	Strong user motivation, immediate value delivery, minimal maintenance burden, achievable timeline.
Overall	RECOMMENDED TO PROCEED	All three dimensions are positive. The project is viable, valuable, and buildable within the given constraints.